

Week 14

Intermediate Machine Learning and Real-World Data Challenges

- Focus; Unsupervised learning, clustering, and tackling incomplete or imbalanced datasets in the context of large-scale AI applications
- Build on; Data ethics, reproducible pipelines, and supervised ML from previous weeks

Learning Objectives

- Understand core clustering algorithms in machine learning
- Recognise and address challenges in working with incomplete or imbalanced data
- Apply techniques to real datasets using industry-standard tools
- Analyze the impact of data scale and complexity on modeling decisions

Agenda

1. Introduction to unsupervised learning and clustering
2. Common clustering algorithms and their applications
3. Real-world data challenges; missing and imbalanced data
4. Practical examples using open datasets
5. Hands-on activity; clustering and data cleaning in Python
6. Industry case studies and discussion
7. Assessment preparation and Q&A

Unsupervised Learning Overview

- Unlike supervised learning, the goal is to find hidden patterns without predefined labels
- Clustering is a main approach to grouping similar data points
- Widely used for exploratory analysis and pre-processing in AI and industry

Clustering Algorithms in Practice

- K-Means; Efficient for large datasets, finds clusters based on distance
- Hierarchical clustering; Builds nested clusters, helpful for understanding data structures
- DBSCAN; Finds clusters of varying shapes, robust to outliers
- Example industries; Retail market segmentation, anomaly detection in finance, image clustering in AI labs

Example; K-Means Clustering in Python

```
from sklearn.cluster import KMeans
from sklearn.datasets import load_iris
import pandas as pd

# Load data
iris = load_iris()
X = iris.data

# Fit K-Means
kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(X)

print(pd.Series(clusters).value_counts())
```

- Try with your exploratory dataset from earlier weeks

Dealing with Incomplete and Imbalanced Data

- Incomplete data; missing entries can distort analysis and model outcomes
- Imbalanced data; classes/groups vastly different in size, affecting predictions
- Real-world scenarios; Health datasets with missing patient info; Fraud detection with few positive cases

Handling Missing Data in Practice

- Removal; Delete rows or columns with missing values (only if justified)
- Imputation; Fill missing values with mean, median, or other calculated values
- Advanced approaches; Use models or data similarity for imputation

```
import pandas as pd
# Fill missing values with column mean
df = pd.read_csv('data.csv')
df.fillna(df.mean(), inplace=True)
```


Addressing Imbalanced Datasets

- Resampling; Over-sample the minority class or under-sample the majority class
- Use of evaluation metrics; Balanced accuracy, ROC-AUC instead of plain accuracy
- Example; In fraud detection, only 1 percent of cases are fraudulent, so resampling or special algorithms are needed

Scale and Complexity in Data Modeling

- Large datasets from open sources can be diverse and complex
- High dimensionality, many missing values, and unequal class distributions are common
- Industrial AI projects rely on scalable, efficient tools and robust cleaning methods

Hands-On Activity; Clustering and Cleaning Exercise

1. Download a UCI ML repository dataset with missing values or imbalanced classes
2. Clean the data using imputation or removal where appropriate
3. Apply K-Means clustering and visualize the clusters
4. Discuss findings; What challenges were encountered? How did cleaning or balancing impact results?

Industry Application; Image Clustering with LAION-5B

- AI research teams cluster images by content to organize and filter large datasets
- Example workflows; Feature extraction with CLIP; clustering similar visuals together to support training LLMs or generative models
- Challenge; Must handle millions of images, incomplete metadata, biased distributions

Group Discussion; Connecting to Workplace Challenges

- Reflect on a workplace or industry scenario where clustering or data cleaning would be critical
- Discuss; What tools or standards (e.g., HuggingFace Datasets, pandas, reproducible pipelines) would you use?
- How would you communicate cleaning steps and quality to non-technical stakeholders?

Assessment Preparation

- Review clustering and unsupervised learning concepts for final project and knowledge test
- Demonstrate handling of missing or imbalanced data in submitted Jupyter Notebooks
- Focus on; Clearly documented cleaning steps, justification of choices, critical reflection on real-world dataset problems

Summary and Next Steps

- Clustering and handling real-world data challenges are foundational for large-scale AI projects
- Apply these methods to open datasets, document your process, and reflect on results
- Next week; Scaling data analysis for very large datasets and advanced AI workflows

References and Further Reading

- "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Géron, Chapter 8; Clustering
- UCI Machine Learning Repository; Explore datasets such as "Adult" or "Credit Card Fraud"
- Scikit-learn documentation; Clustering and Imbalanced-learn
- Responsible AI guidelines; HuggingFace Data Ethics and Documentation Standards

